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Large language models aided patient progression documentation according to the ICD standard☆

Nuria Lebeña [a],*, Arantza Casillas [a], Alicia Pérez [b]

[a] HiTZ Center - Ixa, University of the Basque Country (UPV/EHU). Department of electricity and electronics, Sarriena 2, Leioa 48940, Spain
[b] HiTZ Center - Ixa, University of the Basque Country (UPV/EHU). Department of computer languages and systems, Rafael Moreno "Pitxitxi" 2/3, Bilbao 48013, Spain

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ABSTRACT

Background and Objective Healthcare documentation processing is becoming more and more efficient and effective as a result of advances in machine learning and natural language processing (NLP). One challenge in clinical practice is the early detection of future patient potential diagnoses, which is crucial for preventive medicine. Estimating the potential future diagnoses, helps to speed up the management of Electronic Health Records (EHRs) and opens a path towards clinical prevention. It is a challenging task, as there are thousands of possible diseases, and, in general, there is limited data available to train systems due to privacy concerns.

The objective of his study is to infer future probable diagnoses given patients diagnosis history. In previous works, this task has been carried out using structured data, such as, ICD-coded diagnoses, overlooking unstructured textual information in EHRs. Unlike traditional methods, this study aims to enhance next-diagnosis prediction by integrating patient diagnosis information codified according to the International Classification of Diseases (ICD) with unstructured clinical text.

Methods: We propose a multi-faceted model that integrates structured ICD-encoded patient histories with unstructured EHR text for future diagnosis prediction. Our approach consists of (1) a sequential model trained on structured diagnosis timelines, (2) a Clinical Longformer-based model trained on unstructured EHRs, and (3) an ensemble strategy to combine predictions from both components.

Results: Our proposed ensemble strategy significantly outperforms current state-of-the-art approaches in predicting future diagnoses, achieving a Precision@5 of 72.34% and a Precision@20 of 77.49%. Additionally, it showed high robustness and reliability across different demographic groups and a varying scope of medical history.

Conclusion: This research demonstrates that the integration of structured ICD diagnoses timelines with unstructured EHRs achieves improved results compared to just using structured diagnosis timelines. Notably, the proposed model also maintained high accuracy even with a short-term history of diagnoses.

1. Introduction

The rapid evolution of data-driven approaches in healthcare, particularly through the application of Natural Language processing (NLP) and predictive analytics, has served as the basis for research and practice in medicine. Electronic Health Records (EHRs) are an essential part of research, as they provide relevant information on patients' pathologies. Moreover, they are coded in compliance with International Classification of Diseases (ICD) [1] guidelines that provide a language-independent global coding standard.

Manual analysis of EHRs for patterns is a tedious and costly task that absorbs many healthcare resources. Relieving healthcare workers of this task and allowing them to concentrate on tasks more directly related to the patient is crucial for improving the efficiency of healthcare systems. For this reason, it is imperative to automate the analysis of clinical documentation [2].

The focus of many modern clinical text mining techniques is on preventive medicine, which all together is a branch of medicine that focuses on identifying and reducing risk factors before a new disease

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* Corresponding author.
E-mail address: nuria.lebena@ehu.eus (N. Lebeña).

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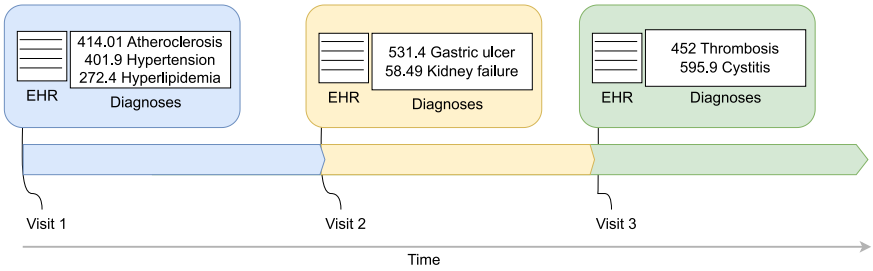


Fig. 1. Illustration of the diagnoses timeline of a patient. This figure shows a series of EHR entries from three different visits, each with associated diagnoses such as atherosclerosis, hypertension, and gastric ulcer. The timeline arranges these conditions chronologically, providing a visual summary of the progression in the health status of the patient over time.

arises or gets worse. This approach aims to reduce the impact of future diseases and thus improve the management of health system resources.

The interest is reflected in new international competitions such as CLEF's Intelligent Disease Progression Prediction 2022, 2023 and 2024 where international research groups participate with the aim of predicting disease progression in a probabilistic, time dependent techniques. The challenge of the competitions focuses on predicting the progression of two severe neurodegenerative diseases: Amyotrophic Lateral Sclerosis and Multiple Sclerosis [3,4]. Moreover, important institutions, such as the European Union, are funding preventive medicine research that focuses on reducing the risk of non-communicable chronic diseases (NCDs) in children and adolescents [5].

While some previous works focus on the integration of clinical and socioeconomic data, such as socioeconomic data, past costs, patient sport habits, the number of previous visits to the doctor, etc. [5,6]. Our research focuses on the automatic analysis of diagnosis timelines for the prediction of probable diagnoses for upcoming admissions, offering clinicians a predictive tool that could potentially improve patient care.

We believe that finding patterns and evolutionary trends in timelines through extensive analysis can help identify possible future diagnoses. The term "diagnosis timeline" refers to the chronology of the diagnoses of a patient in accordance with ICD codes over multiple clinic visits. For each of the visit an EHR captures in an unstructured text what happened during the visit. On the other hand, the diagnoses of the visits are coded according to the ICD and all together make up the diagnoses' timeline. These chronological mappings of medical conditions allow the development of a summary of the expected progression of the health status of a patient. An illustration of a diagnosis timeline is provided in Fig. 1. Note that a patient has an EHR for each visit that accompanies the diagnosis [7].

The aim of our work is to automatically analyze thousands of diagnosis timelines in order to estimate the diagnoses that would arise next. This task is known as "Next diagnoses prediction". Traditional methods primarily rely on structured data, often overlooking the valuable insights hidden in unstructured EHRs [8–10]. Our work addresses this limitation by integrating structured ICD-coded diagnosis timelines with unstructured clinical text. We ensure that no aspect of the medical history of a patient is disregarded, thereby enriching the prediction process with a more complete view of the health trajectory. The challenge of the task is twofold: there are thousands of different ICD codes, making it difficult to analyze diagnosis timelines for predicting future diseases. Moreover, there are few patient EHRs available for research due to privacy concerns, which results in limited data, preventing a comprehensive study of the full range of labels needed for proper research [11,12].

Our study introduces a multi-faceted model designed to tackle the task of subsequent code prediction. This approach combines both diagnostic history in timelines and unstructured text data from EHRs. In an attempt to increase the available data, we propose another preprocessing strategy to expand the size of our training dataset. We trained and evaluated our system under the same conditions as previous works [8–10]. Our approach surpassed the current state of the art approaches, presumably due to the fact that it efficiently incorporated all the evidences of the past evolution of patients to be able to predict future events.

In order to develop a clinical model, it is important to measure its robustness in different aspects, which is why we perform a thorough evaluation of our final model. This entails a demographic analysis to evaluate the performance of the model among various patient groups, looking for possible disparities in predictive accuracy associated with age, gender, or other demographic variables. Moreover, we are concerned with how the effectiveness of our preventive model may decline over time. Therefore, we also investigate how the length of the diagnosis timeline of the patient affects the performance of the system. Notably, our model remains robust across varying patients and also maintained high accuracy even with short diagnoses timelines.

2. Related work

As aforementioned, the task of next admission prediction is of international interest. Current clinical text mining approaches focus on the analysis of disease timelines for the prediction of future diseases.

Research by [13] provides an example of how timelines are used to estimate national health outcomes. With a particular focus on a private Romanian EHR set, their study uses an automated time series machine learning technique to predict monthly counts of new cases for the 10 deadliest diseases according to the World Health Organization (WHO). These predictions are crucial for strategic healthcare planning since they offer pertinent information for the efficient distribution of resources within the Romanian healthcare system. This specific application of the timeline highlights how crucial it is for machine learning tasks aimed at enhancing healthcare system management.

As word embeddings, in recent works, patient embeddings have proven to be effective for summarizing complex and multi-dimensional disease timelines into dense, informative representations. By identifying underlying patterns, these embeddings facilitate a range of tasks including patient categorization, future illness prediction, and treatment pathway discovering. The Time-Aware LSTM (T-LSTM) approach, which improves the creation of patient embeddings for longterm patient records, has made a substantial contribution to this field. Through the addition of a time decay function to the memory cell of the classic LSTM, the T-LSTM provides a way to adapt the influence of past events based on time lapse. This adjustment allows the T-LSTM to generate more accurate and contextually relevant embeddings, which are crucial for subsequent analysis tasks [14].

The study by [15] also aims to infer the next main diagnostic category for seniors with breast cancer. They use the Electronic Health Records (EHRs) diagnosis codes (ICD-9 or ICD-10 ontologies) in a timeline and they developed an LSTM network, which incorporates time decay factors to account for the varying impacts of chronic and acute conditions over time. Additionally, the model employs an attention mechanism to enhance the representation of patient visits by identifying the most influential medical codes.

Building on the increasing interest in using structured timelines of healthcare data for enhanced predictive models, a notable study has leveraged the MIMIC-III dataset to predict future diagnosis codes effectively [9]. This research employs a novel self-attention mechanism that captures contextual dependencies and temporal relationships within a patient's timeline, addressing the intricacies of predicting medical outcomes from EHRs. BiteNet, an end-to-end bidirectional temporal encoder network is able to process and learn from complex, hierarchical patient data structures efficiently.

Another work that addresses the next diagnose prediction task using MIMIC-III; THIGE (Temporal Heterogeneous Interaction Graph Embedding) uniquely combines gated recurrent units (GRU) for capturing short-term patient demands with a heterogeneous self-attention mechanism that embeds long-term patient habits [10]. By incorporating the temporal dynamics of a patient's medical history, this method improves prediction accuracy and demonstrates how embedding techniques can be used to forecast precise patient trajectories.

Transformers, distinguished by their self-attention mechanisms, are particularly adept at modeling complex dependencies without the constraints of sequential data processing inherent to LSTMs and GRUs, allowing them to capture richer, more nuanced representations of patient histories. The recent paper by Blinov and Kokh explores the utilization of transformer-based models for generating patient embeddings, leaving aside the recurrent networks that have traditionally been used for this task [8]. This model not only learns embeddings from the sequential data of patient histories but also integrates demographic factors, enhancing the richness of the embeddings. In addition to providing a condensed view of patient medical histories, these embeddings also improve the explainability of the predictions, as healthcare professionals are able to evaluate the connection between multiple illnesses and their patients.

The aforementioned works have concentrated on the examination of structured data, such as the ICD codes and diagnosis timelines. Although these methods provide a basis for predicting future health outcomes, they sometimes overlook crucial nuances of the unstructured EHR texts that could improve the outcome of the systems.

EHR texts present a challenge as they are unstructured documents that are often incomplete and do not explicitly indicate patient diagnoses. Several studies have addressed data collection challenges and bias elimination in healthcare data [16,17]. Emphasis has also been placed on the importance of preprocessing methods such as imputation, scaling, and feature selection in improving the performance of machine learning models for healthcare predictions [18].

Some works address the next-code predicting task only relying on clinical text. In [19], they propose a novel approach for predicting future diagnoses using discharge summaries from Electronic Health Records (EHR) coming from the MIMIC-III database. They focus specifically on predicting the top 20 most frequently occurring ICD-9 codes, which account for 30% of ICU admissions. They introduce a hybrid deep learning model that leverages a combination of long-short-term (LSTM) networks and convolutional neural networks (CNN).

Nevertheless, a large amount of data is needed to train models that utilize EHRs. One of the challenges today is building a corpus large enough to handle the vast number of ICD codes. Research has focused on different aspects of healthcare data security and interoperability to improve the safety of sharing EHRs and, consequently, build larger datasets [11,20].

Incorporating structured data and EHRs into the models could enhance their performance in the next-diagnosis prediction task. In this work, we propose a multi-faceted model that leverages both diagnosis timelines, which are widely used in patient histories, and EHR texts. In an attempt to increase the available data, we propose another preprocessing strategy to expand the size of our training dataset and make better use of our diagnosis timelines.

Table 1
Summary statistics of the MIMIC-III dataset, detailing the patient demographics in terms of sex and age groups.

		Sex		Total
		M	F	
		55.51%	44.51%	
Age	[0;10)	1.58%	1.45%	3.04%
	[10;20)	0.18%	0.13%	0.32%
	[20;30)	1.25%	1.25%	2.50%
	[30;40)	2.64%	2.14%	4.78%
	[40;50)	6.73%	4.75%	11.48%
	[50;60)	12.60%	8.90%	17.70%
	[60;70)	10.51%	7.20%	2.59%
	[70;80)	12.03%	9.88%	21.92%
	[80;90)	7.95%	8.73%	16.70%

3. Materials

We used the MIMIC-III dataset [7], which consists of 58,976 discharge summaries from the critical care unit of the Beth Israel Deaconess Medical Centre. Clinical documentation services are in charge of saving the discharge summaries together with diagnoses, typically, following the ICD standard [1]. In the case of MIMIC-III, each discharge summary has associated the diagnoses per hospital admission encoded with the ICD-9-CM. The data comprises hospital admissions of 46,520 unique **patients**. The patient demographics details, are presented in Table 1.

Given the close connection of this work with diagnose documentation, we consider it important to devote a short explanation to the standard widely used by clinical documentation services and insurances to encode diagnoses, that is, the International Classification of Diseases, **ICD** [1]. Specifically, we will focus on the ICD version involved in this study, i.e. 9th version. ICD-9 standard encodes diagnostic terms arranged in a hierarchical structure. The hierarchy is illustrated graphically in Fig. 2 and is elaborated in what follows.

In the hierarchy, each **chapter** groups diseases and health conditions based on common characteristics such as their origin or the body system they affect. For instance the chapter Endocrine, Nutritional and Metabolic Diseases relates to conditions that affect endocrine system, as well as nutritional and metabolic functions. Within each chapter, **categories** provide a finer level of detail by further breaking down the diseases into more narrowly defined groups, often specifying the exact nature or etiology of the conditions, as what is, the category related with disorders of Thyroid gland or with nutritional deficiencies. Next, the **main code**, which aligns with the first three digits of the code, identifies a single condition. Goiter is associated with the code 240. Finally, the **fully specified code** includes the three-digit main code followed by a decimal point and one or two additional digits. This extension offers more specific details about the disease, such as its severity and laterality. For example, code 240.0 describes a simple convert case of goiter [21].

We replicated the pre-processing of previous works in order to benchmark against earlier studies [9]. We focused on diagnoses that appeared more than five times and on patients who had more than two visits. This yields 7496 patients and a total of 3874 distinct diagnoses located high up in the hierarchy within 136 categories and 17 chapters. The mean of visits in the resulting set is 2.66. Furthermore, in each visit, several diagnoses (ICDs) are documented; to be precise, in the pre-processed dataset each patient had been diagnosed with between 1 and 266 distinct disorders, the average being 20 diagnoses per patient overall.

Our approach, in contrast to earlier approaches, uses both textual data from the EHRs and traditional medical timeline diagnoses as input to address the problem of next-code prediction. Consequently, we distinguish two distinct types of input data in our work: unstructured EHRs and structured diagnosis timeline data.

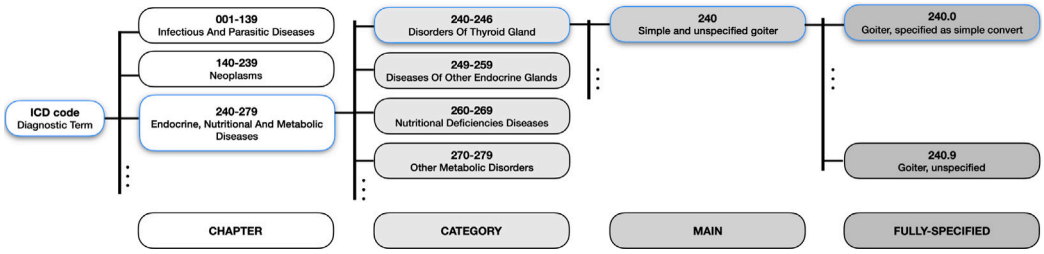


Fig. 2. This figure illustrates the hierarchical Structure of the ICD-9 standard in terms of chapters, categories, main diagnoses and fully-specified diagnostic terms, with each level of the hierarchy involving an increasing degree of detail to define of the diagnostic term.

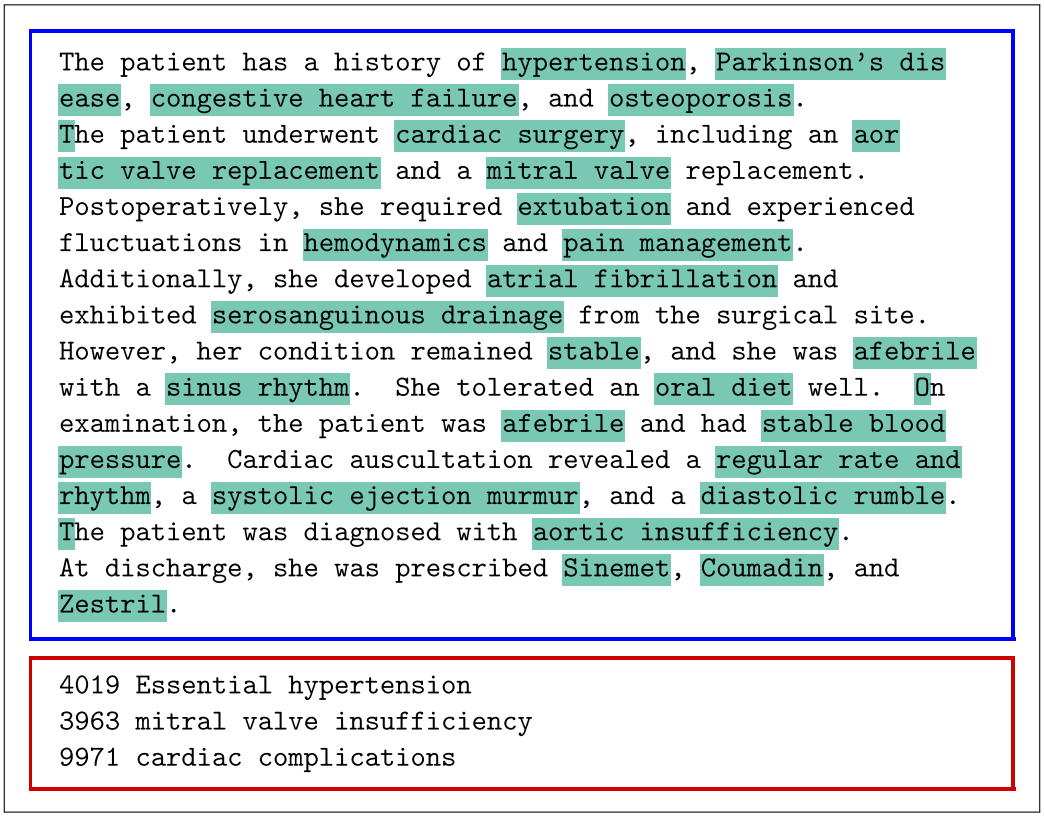


Fig. 3. Example of an EHR input after NER processing and diagnostic information addition. The text framed in blue refers to the EHR, and the information highlighted in green is the one retained after applying NER. The red box shows the associated visit codes and diagnostic terms added.

Table 2
Quantitative description of the EHRs: EHRs represent the total number of documents, the cardinality of the vocabulary and the average number of words per document.

EHRs	26,731
Vocabulary	315,801
Words per EHR	1575

Unlike traditional diagnosis timelines that are sequences of ICD codes chronologically ordered, EHRs are composed of unstructured text. However, they contain valuable information written by physicians that is crucial for improving the classification process. Table 2, shows the amount of EHRs in our dataset, vocabulary size and the average number of words per EHR.

As our focus is on the next diagnosis estimation, we believe it is also important to include the codes and standard definitions of the diagnoses along with the rest of EHR text in order to increase the contextual information. Combining this unstructured information, the added definitions, and the ICD codes themselves is of vital importance, as it completes each patient's record information.

Considering that the length of the base EHRs is large (mean 1545 words) and the capacity of the transformers decreases for large window contexts, we extracted only the relevant entities from the EHRs. In order to accomplish this, we used a named entity recognition (NER) system for biomedical data [22]. The document lengths were considerably reduced by this process.

We use the aforementioned NER system to extract and highlight crucial biomedical entities from the clinical text. We only retain the segments of the document that contain the identified entities as depicted in Fig. 3. The system annotates biomedical and clinical categories such as diagnoses and medical conditions, procedures and interventions, medications, signs and symptoms, and relevant vital signs and observations.

It is important to note that the NER process is applied only to the unstructured EHR text and does not affect the appended ICD codes or diagnostic terms in any way. We then combined in a single text the reduced text of the EHRs (the text highlighted in green in the blue box in Fig. 3) with their associated visit codes, followed by their diagnostic

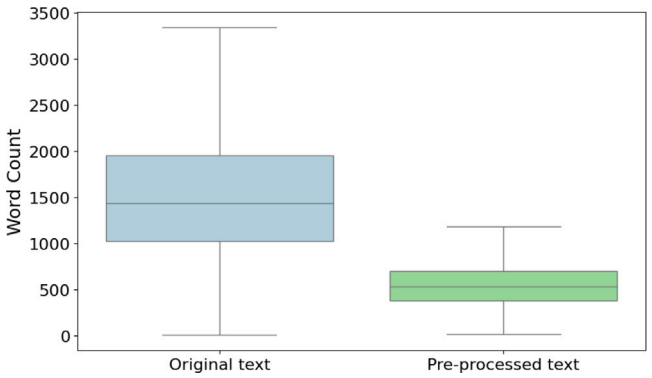


Fig. 4. EHR word counts before and after NER filtering and diagnostic information addition. For both original text and pre-processed text, the chart shows the minimum, mean, mode, standard deviation and maximum word counts.

terms (red box in Fig. 3). This combination enhances the contextual information for diagnosis prediction by providing a comprehensive view of the patient's condition.

Fig. 4 displays a boxplot comparing the word counts of Electronic Health Records (EHRs) before and after the application of NER processing and the addition of diagnostic information. This visualization highlights the reduction in document lengths, with the "Original text" representing the initial word counts and the "Pre-processed text" showing the counts after extracting relevant entities and integrating the codes and their associated diagnostic terms. This text processing results in having more relevant information while considerably decreasing the text length.

3.1. Enhancing inference from partial histories

Unlike previous studies that primarily focused on predicting the outcomes of a patient's final visit (Nth visit) given all history (from visit 1 to $N-1$), our approach modifies this strategy in order to increase the size of our training dataset. By treating each visit separately, we expand our dataset, providing more examples for the model to learn from. We extract 12,506 samples for the original set of 7496 samples.

Fig. 5 provides a comprehensive visual breakdown of how patient data is restructured to enhance the training process for our predictive models. It highlights how we increase the training dataset size by treating each visit as a separate instance. In our case for a visit $N-1$ we use as input codes up to that visit and we predict next visit (Nth visit) codes.

4. Methods

The methodology of our study is based on the use of a multi-faceted model that represents an innovative approach to healthcare predictive analytics. The model offers a thorough analysis of patient evolutions by utilizing both the rich narrative content of EHRs (unstructured data) and the precise diagnostic timelines (structured data).

Fig. 6 presents a schematic of our model. The framework consists of two different data inputs: on the one hand the structured diagnostic timelines, and on the other hand the unstructured preprocessed EHRs. To effectively process and analyze each data type, our combined model consists of two specialized main approaches. While the first one focuses on extracting the next diagnosis from the structured diagnostic timelines, the second one is in charge of deriving the next diagnosis from the unstructured EHRs.

In this section we look into the insights of each approach. Section 4.1 discusses the approach for estimating next diagnoses given diagnose-timelines. Section 4.2 describes the method used to forecast future diagnoses with unstructured EHRs as input. Finally, Section 4.3 details how we integrate the predictions from both components of our multi-faceted model. The details of the software and hardware requirements have been included in Appendix A. Software and Hardware Requirements.

4.1. Estimate next diagnoses given the diagnose-timeline

Given a patient timeline history, it is important to be able to estimate the subsequent diagnoses to reinforce clinical prevention. Disease timelines themselves provide rich, structured information that is used to analyze potential co-relationships between diseases and establish potential future diagnoses.

We first focused on analyzing methods to classify structured diagnosis timelines. While previous works focus on using strategies that take into account temporal dynamics such as bidirectional temporal encoder networks [9] or gated recurrent units [10], training transformers from scratch has shown very competitive results with respect to previous methodologies [8], improving the state-of-the-art.

Consequently, as a base approach we first wondered if the outcome would improve by using bigger transformers networks. Unlike previous work that restricted their transformer structure to 4 encoder layers, 4 self-attention heads and an embedding space of 256 dimensions [8], we used a model with the same structure as the BERT-large model [23], consisting of 24 transformer layers, 1024 hidden units per layer, and 16 attention heads. We referred to this base approach as "**Sequential scratch**".

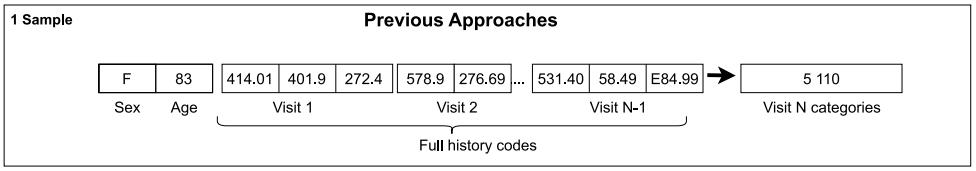
In recent years, there has been a trend to use pre-trained LLMs that have a broad contextual knowledge of different knowledge areas (medicine, law...) or languages (pre-trained BERT with Spanish texts [24], multilingual BERT [25]...). By using these pretrained models, researchers can take advantage of rich pre-existing knowledge, significantly reducing the time and resources required for training transformers from scratch. LLMs adapted to the clinical domain trained with a large corpus of medical and clinical texts, enable the understanding and generation of relevant information in healthcare contexts. The purpose of the study was to investigate whether an LLM adapted to the clinical domain could improve performance in a data-poor setting like ours, where the main data elements are structured codes instead of traditional text. Therefore, we next wondered about the potential benefits of using a pre-trained large clinical language model (clinical-LLM) [26]. We integrated all of the ICD-9 codes into the vocabulary of the transformer. We first pre-trained and fine-tuned clinical-LLM with the new vocabulary data, thus creating our proposed "**Sequential-pretrained**" model.

The performance of our "Sequential scratch" and "Sequential-pretrained" are presented in section Section 5

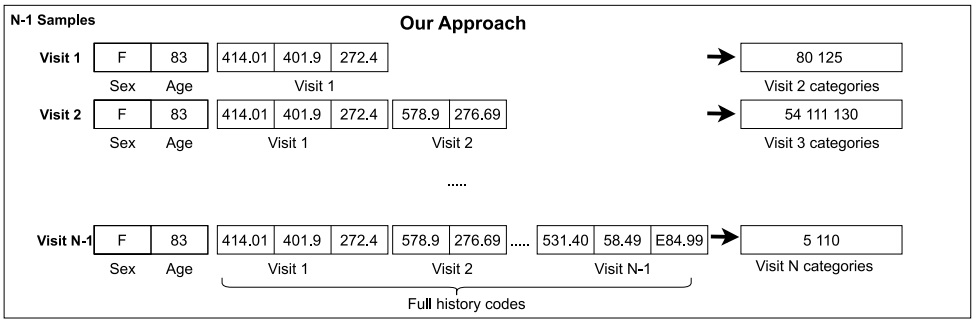
4.2. Estimate next diagnoses given unstructured EHRs

To effectively process the unstructured information found in EHRs, we employ the Clinical Longformer [27] model, a variant of the Longformer [28] specifically adapted for clinical documents. The Longformer's architecture is specifically designed to manage extensive texts efficiently, making it particularly suitable for the detailed and expansive narratives contained in EHRs. This is critical as EHRs typically contain much more than the 512-token limit imposed by traditional transformer models such as BERT [23], which significantly constrains their ability to handle lengthy medical documents without losing essential information.

As stated in Section 3 we filtered EHRs using NER and added diagnostic information. This resulted in an average length of 562 words per document. However, this still meant that only about 46% of documents could be processed in their entirety under the 512-token limit, leading to potential loss of critical contextual information. The Clinical Longformer model, which extends the token capacity to 1024, allows us to handle over 95% of EHR documents without the need for truncation.



(a) Previous methods predict the outcome of the final visit of a patient using all historical data from previous visits



(b) In our approach the training dataset is increased by treating each visit as an independent training instance enabling next disease prediction

Fig. 5. Strategic restructuring of patient data to optimize the training process for our predictive models. This diagram contrasts our approach with previous methods, which typically focused on predicting the outcome of a patient's final visit using historical data from all prior visits. Our approach increases the training dataset by treating each visit as an independent training instance, using the codes up to the $N-1$ visit as input to predict the codes for the Nth visit. For simplicity, we have illustrated the categories with numbers instead of code ranges, but each category is associated with certain diagnostic-term codes, as illustrated in Fig. 2.

4.3. Enhance inference with prediction combination strategies

In previous sections, we have discussed the main approaches to classify structured diagnosis timelines and unstructured EHRs:

1. *Processing structured diagnosis timelines*: We proposed a robust model able to deal with sequential codes and produce next diseases.
2. *Processing unstructured EHRs*: We presented a model that can handle longer unstructured text: the "clinical-longformer" model.

The outputs from these models are essentially the confidence each model assigns to various clinical categories. These outputs are then combined through various statistical aggregation techniques:

- Maximum: This method selects the highest confidence value of the confidence scores produced by both models for each possible clinical category.
- Minimum: Conversely, this method chooses the lowest confidence value.
- Weighted Average: Computes the mean weighted by the precision of each model.

Through the use of these aggregation techniques, our model makes use of the advantages that each component offers in order to produce more accurate predictions.

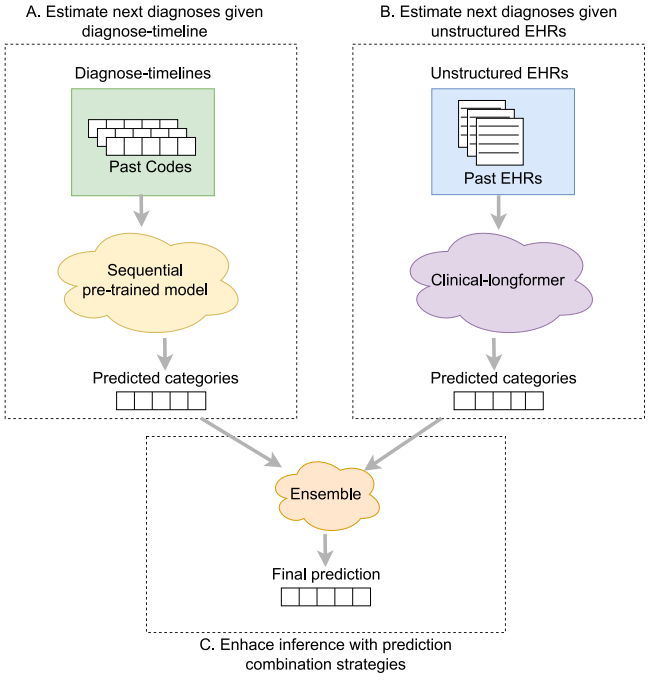


Fig. 6. Schematic diagram of the multi-facet strategy framework. This graphic illustrates the main approaches of our combined model: first one to *estimate next diagnoses given diagnose-timeline*; second one to *estimate next diagnoses given unstructured EHRs*. Moreover, it illustrates how the two models work together to *enhance inference with prediction combination strategies*.

5. Experimental results

In this section, we present the findings of our multi-faceted model. We conduct a series of evaluations designed to assess the robustness and versatility of our approach under various scenarios. As aforementioned, we employ our increased training dataset as depicted in Fig. 5(b).

We begin by comparing the results of our "**Sequential scratch**" and "**Sequential pretrained**" models in the task of estimating the next diagnoses given the diagnose-timeline.

We then introduce the different ensemble methods. Note that antecedents merely focus on the estimation of the diagnosis inferred for the last visit of the patient given the full history in a timeline. Therefore, we next determine the performance of our combined model using full history timeline diagnosis (Fig. 5(a)) sequences in order to benchmark our multi-faceted model against antecedents. These results are included under Section 5.2. In Section 5.3, we thoroughly assess the robustness of the model across different experiments. We first analyze how much historical data is required for optimal performance by assessing the effect of the number of visits on prediction accuracy.

Table 3
Performance in the last diagnoses prediction task compared with state of the art.

System	Precision@k	
	k = 5	k = 20
MusaNet [10]	65.07	71.04
BiteNet [9]	66.15	71.04
MedicalProfileModel [8]	66.77	72.58
Sequential-scratch	66.48	71.50
Sequential-pretrained	**68.31**	**74.10**

Then, in Section 5.4 we provide the results in demographic groups to ensure consistent and equitable predictions.

5.1. Estimate next diagnoses given the diagnose-timeline

As stated in Section 4.1 on this first experiment, we evaluate the performance of our models aimed at classifying structured diagnosis timelines: "**Sequential scratch**" and "**Sequential pretrained**".

We follow the tendency of previous work and merely focus on the estimation of the diagnoses inferred for the last visit of the patient given the full history in a timeline. We therefore use our diagnose timelines structured as depicted in Fig. 5(a) as previous works did [8–10].

The experimental results are shown in Table 3, for comparison purposes, as in previous works, we used a 10-fold cross validation to estimate the mean of Precision@k for k = [5, 20] values [8–10]. We trained our "Sequential-scratch" model for 50 epochs with a batch size of 32. As demonstrated in Table 3, although this strategy yields results slightly below those reported by [8], it surpasses the outcomes achieved by [9,10]. This indicates that transformers are an effective option for performing the task.

The "Sequential-pretrained" model, which has already been pre-trained with clinical information [26], was further pre-trained with our training data using the Masked Language Modeling (MLM) task, also with a batch size of 32 and for 50 epochs. The results shown in Table 3 led us to conclude that the "Sequential-pretrained" model, which made use of the clinical-LLM model, produced better outcomes. This resulted in its incorporation into our final multi-faceted model, which provides improved predictive accuracy in sequential code classification by utilizing the advantages of an LLM with previous clinical knowledge.

5.2. Predicting next diagnoses

In this section we assess the ability of our approach, depicted in Fig. 6, to predict next diagnoses. We assess each of the models involved (sequential pre-trained and clinical-long former) to get information from either sequential time-lines or raw EHRs; finally, we assess the ensemble incorporating alternative combination strategies.

We implemented three traditional ensemble strategies: max ensemble, mean ensemble, and weighted ensemble. These methods integrate the predictions from both systems, determining the final prediction by either taking the maximum value, the arithmetic mean, or a weighted mean of the initial predictions. In the weighted ensemble approach, weights were assigned based on the performance of each system using the precision@10 metric, where 10 corresponds to the average number of labels per data instance.

Precision metric measures the percentage of categories that were correctly inferred by the system within the predictions that obtain a higher score than the threshold. We evaluated our model's precision across decision thresholds from 0 to 1, in 0.1 increments. This analysis helps understand the model's performance under varying prediction acceptance criteria. Fig. 7 shows precision results for each threshold, allowing for optimal threshold selection. We observe that the ensemble improves the outcomes when compared to using the models separately.

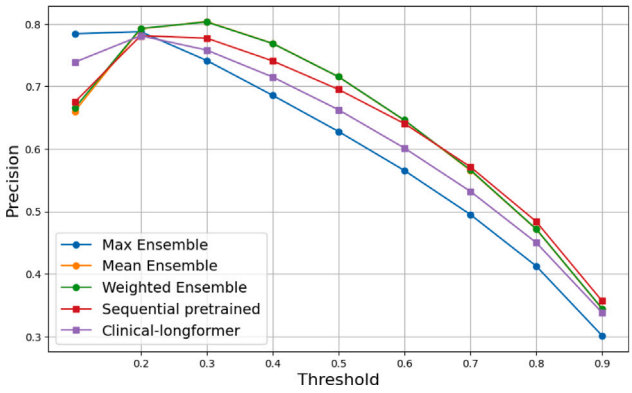


Fig. 7. Precision of the system with augmented patient histories in the next diagnoses prediction for thresholds ranging from 0.1 to 0.9.

Table 4
Performance with augmented patient histories in the next diagnoses prediction task for $K \in \{1, 5, 20\}$.

Data-input	System	Precision@k		
		k = 1	k = 5	k = 20
Diagnosis timelines	Sequential pre-trained	89.24	71.94	76.6
Unstructured EHRs	Clinical-longformer	88.62	67.98	73.93
Both data types parallel input	Ensemble-mean	**90.87**	72.15	77.44
	Ensemble-max	89.78	71.77	77.09
	Ensemble-weighted	90.85	**72.34**	**77.49**

While Fig. 7 shows precision across various thresholds, we also evaluate our models using threshold-independent metrics. For the sake of comparison with previous works, we have evaluated the performance of the original models and ensembles according to p@1, p@5, and p@20. Results are presented on Table 4.

5.3. Prediction robustness

We delve into the robustness of our predictive model by examining its performance across different scenarios to understand its reliability and consistency. Given a patient history up to visit N, it is important to be able to estimate the subsequent diagnoses to reinforce clinical prevention. We hypothesize that estimating next diagnoses with little information about a given patient, that is, estimating with little context or history might be difficult. Therefore, we first assess the performance of the model relative to the length of their diagnoses timelines. We analyze the number of previous diagnoses required to correctly predict the categories of the next admission of a patient. In order to do this, we concentrated on the sequential-pretrained model and measured the predictive accuracy of the model at each sequence length by feeding it the first N codes. For example, for sequence length 10, we cut all patients' timelines up to the 10th diagnosis. We then try to predict the next admission diagnosis using this input and evaluate the system.

The results, depicted in Fig. 8, show that the predictive capacity of the model increases with the sequence length. This implies that our model's ability to predict a patient's next category with greater confidence increases with the number of sequence codes available. Therefore, compared to patients with limited available data, our system can better determine the next diagnosis for patients with multiple historical entries. In other words, the more history we can count on, the better the precision of the next diagnosis.

5.4. Age and gender bias

We now move on to question how this aspect is impacted by demographics. Is the likelihood of receiving the correct future diagnosis

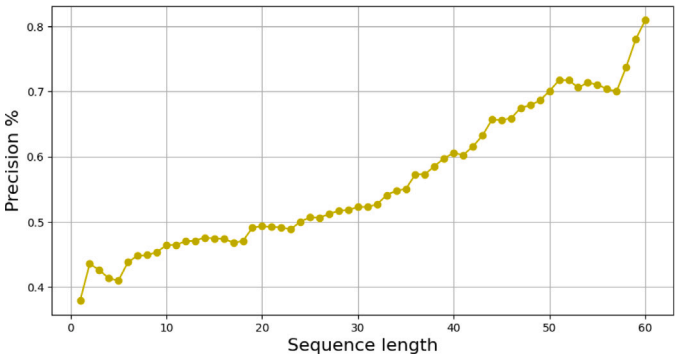


Fig. 8. Predictive performance as a function of diagnoses sequence length.

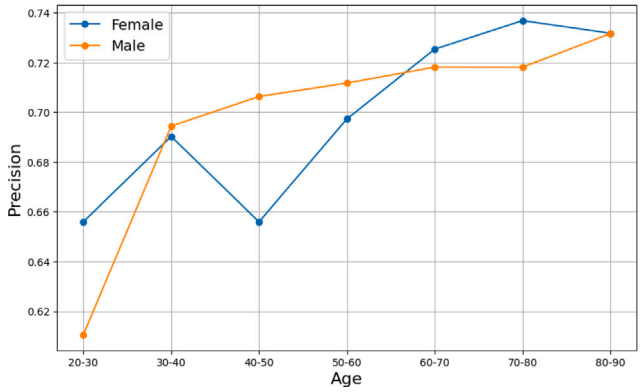


Fig. 9. Precision of predictive performance by age and gender. This graph illustrates the variation in prediction precision across different age groups for male and female patients. Relevant statistical information can be found in Table 1.

higher for older patients? What about the gender bias? In an attempt to illustrate this, we split the patients by sex and age and calculated the precision by age range. We show these results in Fig. 9. It can be seen that for both genders, the tendency to increase precision with age is maintained.

From the results in Fig. 9, we derive that our system tends to produce more accurate predictions as patients get older. Arguably, it is true that the older the patient, the more extensive the clinical history and, as shown in Fig. 8, a lengthy history facilitates the task of identifying future disease risks.

In this regard, we discovered minor variations in the future diagnoses predicted by gender, observing a minor decline for women in the 40–50 age range and a minor increase for those in the 60–80 age range. This is essential to guarantee that the application of these models in the healthcare industry does not introduce any bias.

6. Discussion

The development of our predictive model relied on the premise that EHRs texts contain patterns that can aid in predicting future medical conditions. By integrating structured ICD-coded data with unstructured EHR text, our model leverages the complementary nuances offered by both data types.

We evaluated our model in a variety of scenarios: predicting next diagnoses, evaluating the results according to patient demographics, and evaluating prediction robustness. We paid attention to differences in patient demographics and their diagnosis timelines. First, the integration of structured diagnosis timelines with unstructured EHR text significantly improved predictive performance. Our ensemble model improved up to 2% the results obtained by training the models separately. If we compare this outcome to previous work [8], the improvement is of 5%. Our results show that the predictive capacity of the model increases with sequence length, and it is more complicated to determine the future diagnosis of patients for whom the known past history is small. This is also in line with the clinical vision of healthcare professionals, for whom it is also difficult to determine potential complications in patients with few pathologies or without a complete diagnosis. Furthermore, the demographic analysis shows that no particular patient group is disproportionately better or worse classified. Gender and age bias analysis is important to be aware of the limitations of the approaches proposed.

We have made significant progress in predicting future medical conditions with our multi-faceted model, which integrates both coded entries and unstructured text from EHRs. Despite the model's promising performance, several challenges were encountered during its development. Data availability remained a significant hurdle, as privacy regulations restricted access to larger and more diverse datasets, limiting the model's generalizability. Additionally, the complexity of the ICD code hierarchy made it challenging to identify meaningful patterns without substantial data. The training of the Clinical Longformer required significant computational resources, and even after NER-based text reduction, some documents still exceeded the model's token limit.

In this paper we focus on ICD-9 but our work is not limited to this version of the ICD. Importantly, our approach is designed to be completely general and can be adapted to any revision of the International Classification of Diseases, including the latest ICD 10th revision. This versatility ensures that our model does not depend on the specific characteristics of any ICD version, enabling its application across future updates.

Furthermore, our model is also adaptable to other languages. This feature is essential for extending the use of our approach in health systems with different official languages.

7. Conclusion

The aim of this work is to tackle the "Next diagnoses prediction" task, by automatically analyzing thousands of diagnosis timelines to estimate the diagnoses that would arise next.

Several related research efforts in next-diagnosis prediction have faced significant challenges, including limited data availability due to privacy concerns and the complexity of managing thousands of ICD codes. Models like T-LSTM, BiteNet and Transformers have provided innovative solutions for handling temporal dependencies and patient embeddings; achieving competitive results in the task [8–10]. Nevertheless, they often overlook the information contained in clinical text.

Our contribution rests in the use of a multi-faceted model that captures a wider range of patient information, including EHRs textual notes often overlooked in the task.

Through this approach and the effective use of a distinct patient data structure, results reveal that our system is able to be robust to natural variations present in real clinical data. Moreover, the demographic analysis incorporated into our research shows that the model works better in older patient groups, underscoring our commitment to developing fair and unbiased AI tools for healthcare.

This research provides a valuable tool for healthcare professionals, aiding in the identification of potential future conditions and improving patient care. Moreover, in healthcare administration, this approach can enhance resource allocation by predicting patient admission rates and potential diagnoses, ensuring that hospitals and clinics are better prepared to meet patient needs.

Beyond the healthcare sector, the insurance industry can leverage our approach to forecast future claims and optimize premium calculations based on anticipated medical conditions. Additionally, the pharmaceutical industry can benefit from disease forecasting to refine

drug development processes and improve patient selection for clinical trials.

Extending the outcome to predict not only the diagnoses a patient might receive in the future but also the timing of those diagnoses is an exciting area for future research. The time gaps between patient visits, are not taken into consideration by current models. Predicting the timing of a future diagnosis could significantly enhance the efficacy of the models, providing more accurate insights for clinical decision-making.

Another future task is to test the model presented in this research on textual data in languages other than English, such as Spanish. The objective is to quantitatively measure the model's adaptability to other languages.

Moreover, as the demand for transparency in AI systems increases, especially from large institutions, it is crucial to incorporate explanations into our predictions. Future work could focus on providing dual explanations for the developed multi-faceted model.

Firstly, patient embeddings from the sequential model could be extracted to provide statistics or information from similar patients. Secondly, methods to identify important words or phrases in the document that influenced a prediction could be employed.

CRediT authorship contribution statement

Nuria Lebeña: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Arantza Casillas:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Funding acquisition. **Alicia Pérez:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Software and hardware requirements

A.1. Software dependencies

For the implementation of the Sequential pre-trained model and the Clinical longformer model, we employed Python 3.9.7 programming language next to the Transformers version 4.27.3 - Hugging Face's library:

- Our Sequential pre-trained model is pre-trained on top of the Hugging Face's UFNLP/gatortron-base model [26].
- We use the Hugging Face's yikuan8/Clinical-Longformer [27] model for text processing.

A.2. Hardware requirements

The experiments were conducted using an NVIDIA Tesla V100 GPU (32 GB) and 138 GB of CPU.

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